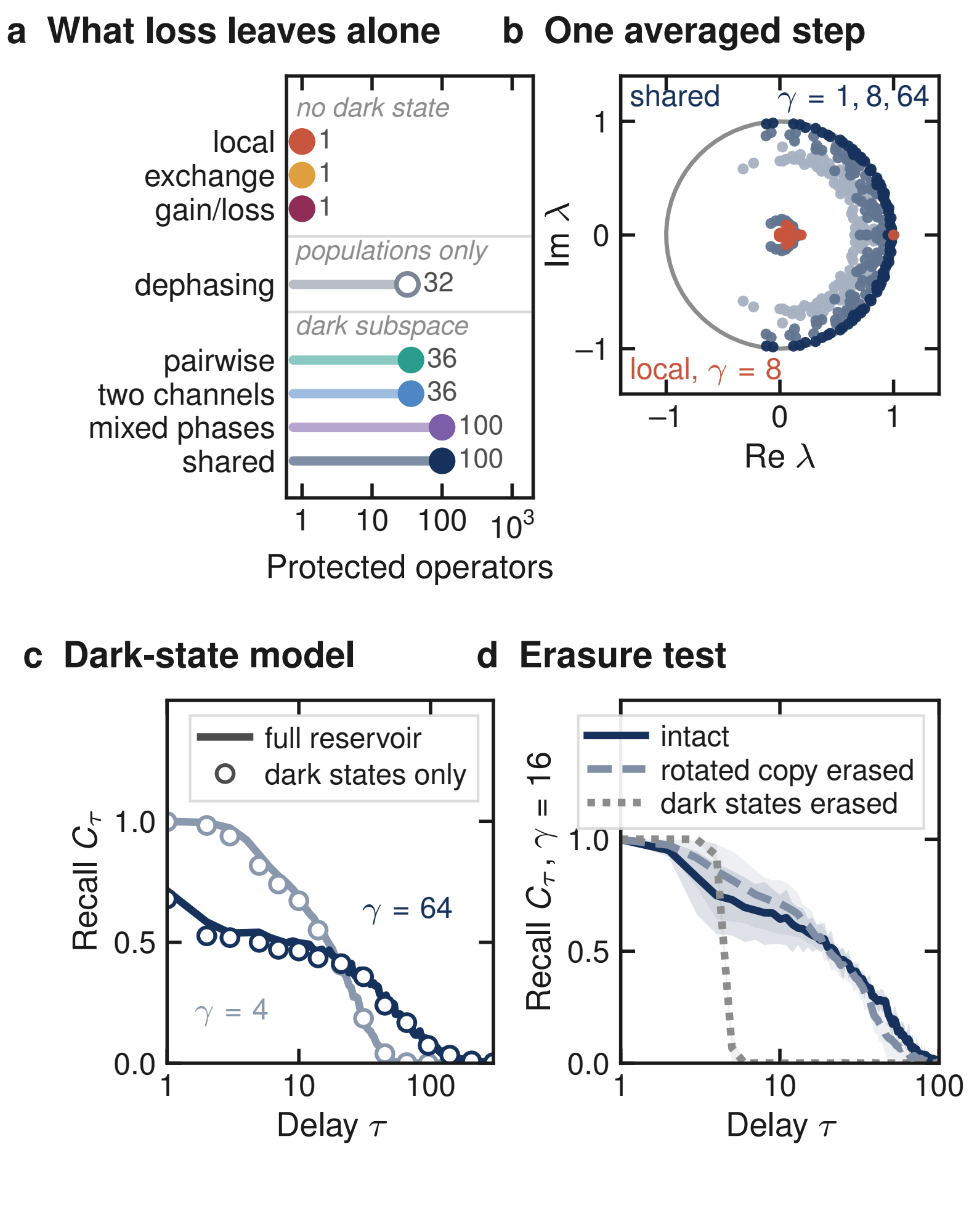

a What loss leaves alone
no dark state
local
exchange
gain/loss
1
1
1
populations only
dephasing
32
dark subspace
pairwise
two channels
mixed phases
shared
36
36
100
100
1
10
100
10^3
Protected operators
b One averaged step
shared
γ = 1, 8, 64
local, γ = 8
Im λ
Re λ
1
0
−1
−1
0
1
c Dark-state model
full reservoir
dark states only
γ = 64
γ = 4
Recall C_τ
1.0
0.5
0.0
1
10
100
Delay τ
d Erasure test
intact
rotated copy erased
dark states erased
Recall C_τ, γ = 16
1.0
0.5
0.0
1
10
100
Delay τ